

Dipayan Sarkar

Siliguri, West Bengal, India

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Summary

Works on de novo protein design with deep generative models - autoregressive and diffusion-based approaches over biological sequence data. Takes models end-to-end, from architecture design and large-scale GPU/TPU training through deployed, production inference platforms.

Skills

Deep Learning: Transformers, attention mechanisms, transfer learning, sequence modeling
Generative AI: Autoregressive, diffusion-, and flow-based generative models for de novo protein design
Programming: Python (PyTorch), R (published CRAN package), C
Scalable Training: GPU-based training, TPU (v4/v5e/v6e), parallel training on HPC environments
Deployment: Docker, Flask, FastAPI, Linux server deployment, model-backed web applications
Data & Viz: NumPy, Pandas, Matplotlib, Seaborn
Domain: Biological sequence analysis, protein-protein interaction prediction, biomolecular network analysis

Experience

University of North Bengal <i>Machine Learning Researcher (PhD Research Scholar)</i> Department of Bioinformatics	Siliguri, India <i>2022–Present</i>
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- Deep learning architectures for sequence-based protein-protein interaction prediction using attention-based hybrid architecture with transfer learning; published in *BBA - Proteins and Proteomics*.
- Sequence based generative modeling of protein binder generation using autoregressive and diffusion-based approaches, including a sparse mixture-of-experts routing framework.
- Development and production deployment of web-based platforms for model inference and protein interaction network visualization.
- Secured competitive compute grants (IndiaAI NVIDIA L4 cloud, Google TRC TPUs) and ran training workloads across both.

Software & Tools

Deep-Interact Studio: Interactive web platform for building, training, and comparing custom deep learning models for protein-protein, drug-target, RNA-protein, and protein-DNA interaction prediction, with integrated interpretability.
deepinteract.compbiosysnbu.in/

AttnSeq-PPI: Deep learning-based protein-protein interaction prediction platform.
compbiosysnbu.in/attnseqppi/

EGRNi: R package for ensemble gene regulatory network inference, published on CRAN.
cran.r-project.org/package=EGRNi

Education

University of North Bengal <i>Ph.D. in Bioinformatics</i> Deep learning-based frameworks for protein-protein interaction network prediction. Advisor: Dr. Chiranjib Sarkar.	India <i>Expected March 2027</i>
University of North Bengal <i>M.Sc. in Botany (Biochemistry), 77.25% - First Class</i>	India <i>2021</i>
Ananda Chandra College, University of North Bengal <i>B.Sc. (Hons.) in Botany, 63.50% - First Class</i>	India <i>2019</i>

Selected Publications

2025: AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *BBA - Proteins and Proteomics*. doi:10.1016/j.bbapap.2025.141102

2026: MoE-Bind: Guiding de novo protein binder generation with sparse experts. *bioRxiv* preprint. doi:10.64898/2026.06.13.732043

2026: ARACoFusion: Uncertainty-aware calibrated deep learning for protein-protein interaction network prediction. *bioRxiv* preprint. doi:10.64898/2026.05.22.727120

2026: Deep-Interact Studio: An interactive deep learning model building platform for biomolecular interaction prediction. *bioRxiv* preprint. doi:10.64898/2026.07.02.736034

Awards & Compute Grants

2025: IndiaAI Compute Initiative - NVIDIA L4 GPU cloud award under the IndiaAI Mission (Project ID: P1-S2025070964, Sep 2025 – Aug 2026).

2025: Google TRC (TPU Research Cloud) Award - Free access to Google Cloud TPUs (v4, v5e, v6e) for machine learning research.

2021–22: CSIR-NET (JRF) in Life Sciences - **All India Rank: 216**; **GATE** in Life Sciences (XL) - Qualified